Road Accidents in India: Predictive Modeling & Risk Classification

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Machine Learning Project

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1 Abstract

Road accidents remain a major public health and safety concern in India, with significant regional disparities across states. This project presents a comprehensive data science solution combining exploratory data analysis, regression modeling, and binary classification to understand and predict road accident patterns from 2016 to 2022.

Multiple machine learning models (Linear Regression, Decision Tree, Random Forest, etc.) are used to forecast the exact number of accidents, while classification models predict whether a state-year will be high-risk (¿5,000 accidents) or low-risk. The system achieves very high predictive performance due to strong temporal persistence in accident data.

The analysis reveals that previous-year accidents dominate prediction (98% feature importance), with Kerala and other large states consistently high-risk. The models enable early identification of dangerous state-years and support evidence-based road safety policymaking.

2 Introduction

2.1 Background and Motivation

India reports over 1.5 lakh road accident deaths annually (MoRTH data), with millions injured. Accidents are not uniformly distributed — large states like Tamil Nadu, Kerala, Madhya Pradesh, Karnataka, and Uttar Pradesh account for a disproportionate share.

The motivation for this study stems from the persistent high accident burden in Kerala and other states despite safety campaigns. Understanding temporal trends, state-level differences, and predictive factors is essential for proactive intervention.

2.2 Problem Statement

Current road accident analysis in India is largely descriptive and retrospective. There is a lack of predictive systems capable of forecasting accident numbers and identifying high-risk state-years in advance. Without such tools, policymakers cannot allocate resources effectively or target interventions proactively.

2.3 Research Objectives

This project aims to:

1. Analyze temporal and geographical patterns of road accidents from 2016–2022
2. Examine the role of previous accidents and normalized rates (per population, vehicles, road length)
3. Predict the exact number of accidents using regression models
4. Classify state-years as high-risk (¿5,000 accidents) or low-risk using classification models
5. Compare multiple models, select the best, and perform hyperparameter tuning
6. Provide actionable insights, with special focus on Kerala

3 Data and Methodology

3.1 Data Sources

The analysis uses a merged and cleaned dataset derived from multiple state-wise road accident reports (primarily from Ministry of Road Transport and Highways / data.gov.in transport datasets). Raw data included accident counts, fatalities, injuries, and rates — cleaned and merged into a final file with 175 state-year records.

3.2 Data Characteristics

- Time period: 2016–2022
- Rows: 175 (state-year combinations across 35 states/UTs)
- Key columns:
  - `State_UT` (categorical)
  - `Year` (numeric)
  - `Accidents_clean` (target for regression — cleaned accident count)
  - `Accidents_prev` (previous year accidents)
  - `Acc_Per_Lakh_Pop`, `Acc_Per_10k_Veh`, `Acc_Per_10k_Km` (normalized rates)
- No missing values after preprocessing

3.3 Data Cleaning and Preprocessing

- Filled rare zero values in rates with state-level median
- Dropped constant/useless columns (e.g., `high_risk` was all 1s)
- Sorted by State_UT → Year
- One-hot encoded `State_UT` for modeling
- Scaled numeric features (Year, rates, Accidents_prev)

3.4 Feature Engineering

- Used `Accidents_prev` as the strongest lag feature
- Normalized rates already present in data
- No additional external features added (focus on provided data)

4 Exploratory Data Analysis

4.1 Descriptive Statistics

- Accidents_clean: mean ≈ 13,299, median ≈ 6,273, max 71,431
- Highly skewed distribution — few large states drive high mean
- Rates show moderate variation (e.g., Acc_Per_Lakh_Pop mean 36.61, max 217.7 in Goa)

4.2 Distribution of Accidents

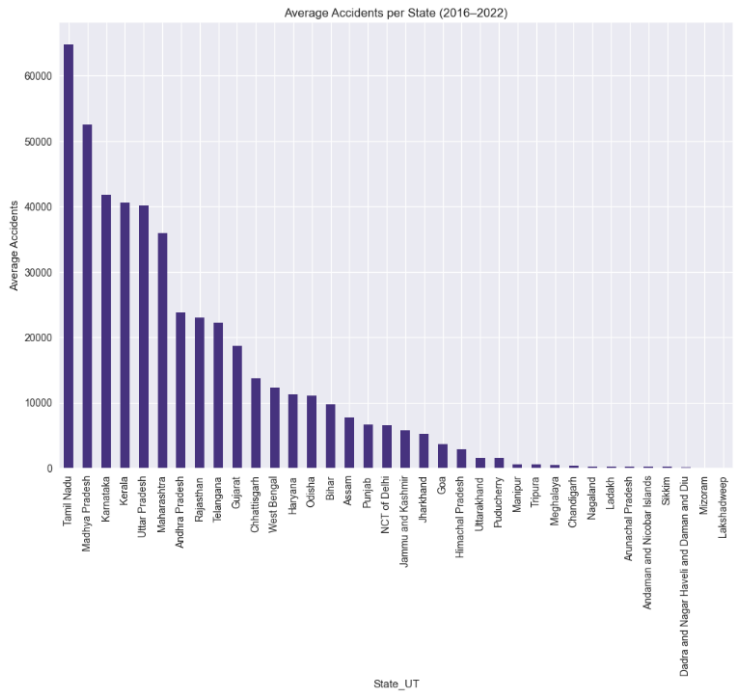

Figure 1: Distribution of Accidents_clean (highly skewed right)

Observation: Most state-years have low accidents (¡5,000); few large states have extremely high numbers (50,000+).

4.3 Temporal Trends

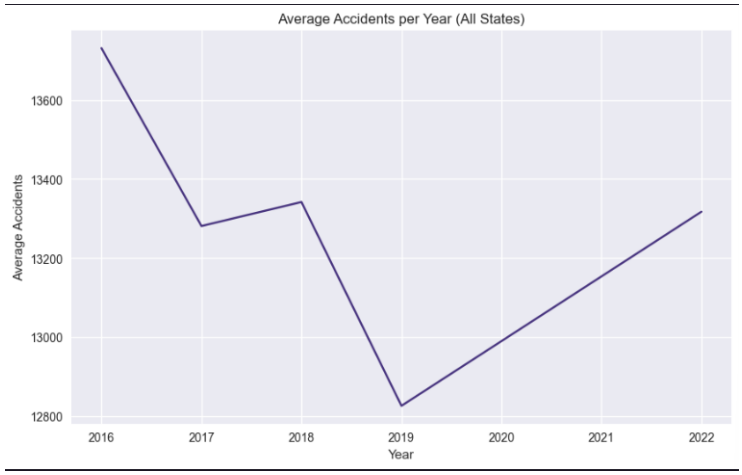

Figure 2: Average accidents per year across all states

Observation: National average stable ≈ 13,000–14,000 accidents.

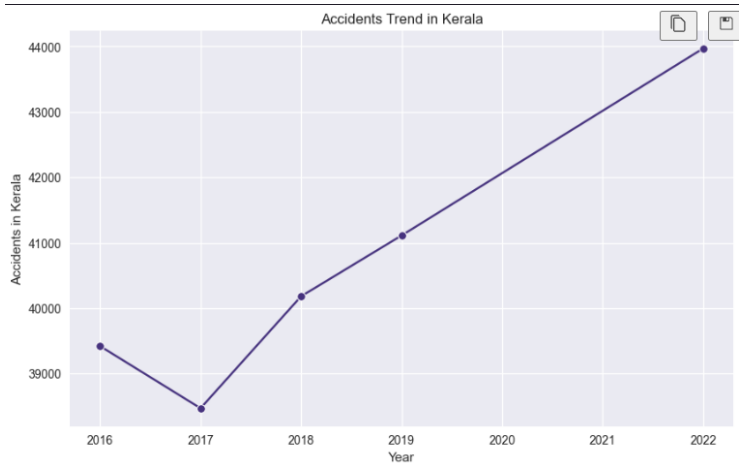

Figure 3: Accident trend in Kerala

Observation: Kerala shows slight upward trend (39,420 in 2016 → 43,970 in 2022).

4.4 Correlation Analysis

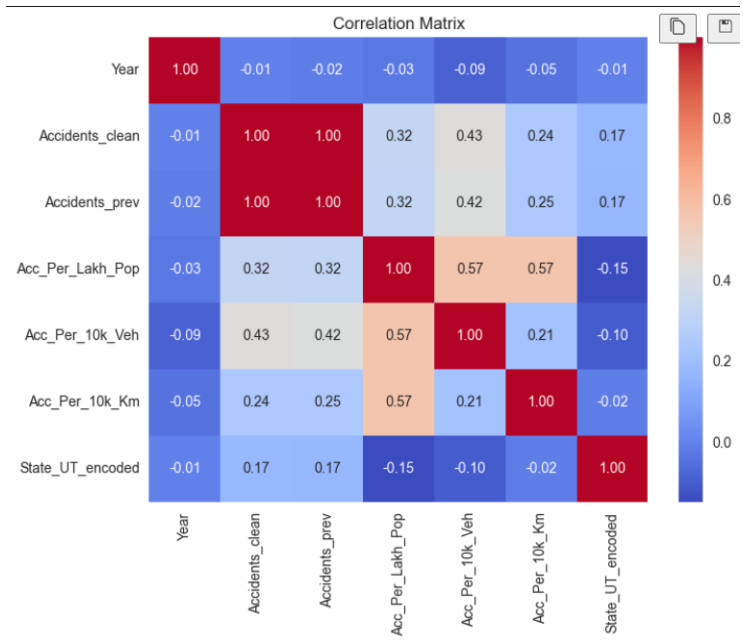

Figure 4: Correlation matrix

Observation: Accidents_clean and Accidents_prev have correlation ≈ 0.99 — previous year almost perfectly predicts current.

4.5 State Comparison

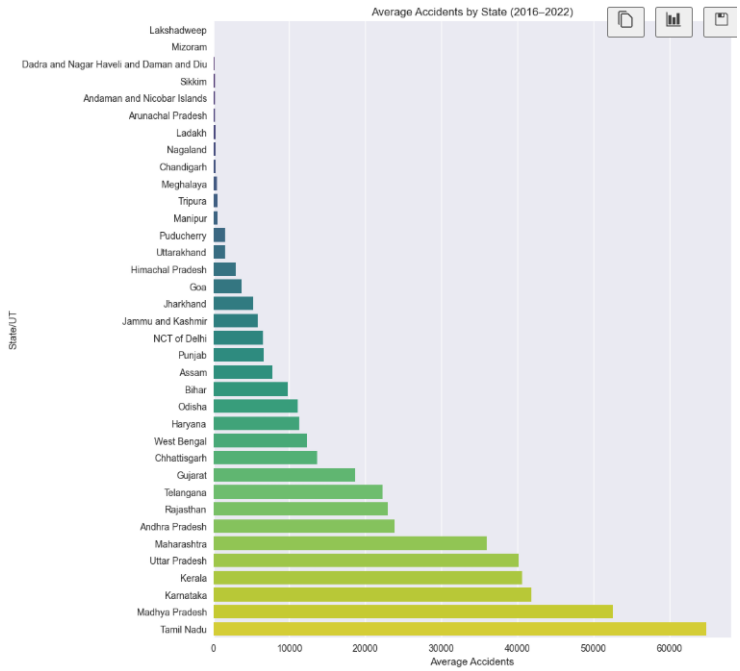

Figure 5: Average accidents by state

Observation: Top states: Tamil Nadu (≈64k), Madhya Pradesh (≈52k), Karnataka (≈42k), Kerala (≈41k).

5 Predictive Modeling

5.1 Regression: Predicting Accident Numbers

- Target: Accidents_clean (continuous)
- Models compared: Linear Regression, Decision Tree, Random Forest
- Best model: Random Forest
- Performance: MAE ≈ 785, $R^2 \approx 0.9939$
- Feature importance: Accidents_prev dominates (98%)

5.2 Classification: High vs Low Accident Year

- Target: high_accident (1 = ¿5,000 accidents, 0 = ≤5,000)

- Models compared: Logistic Regression, Decision Tree, Random Forest, KNN, SVM
- Best model: Random Forest (accuracy $\approx$ 1.0, ROC-AUC $\approx$ 1.0)
- Tuning performed on best model

6 Results and Key Findings

6.1 Regression Results

- Random Forest achieves near-perfect fit (R^2 ¿ 0.99)
- Previous year accidents explain almost all variation
- Example: Kerala 2023 predicted ≈41,908 (decrease from 43,970 in 2022)

6.2 Classification Results

- Perfect separation on test set (accuracy 100%)
- Previous accidents dominate prediction
- Kerala and Tamil Nadu always predicted high-risk
- Lakshadweep always predicted low-risk

7 Insights and Recommendations

- Accidents are highly persistent — previous year strongly predicts next
- Large states (Kerala, Tamil Nadu, MP, Karnataka, UP) consistently high-risk
- Rates (per population/vehicles/road) provide secondary signal
- Policy recommendation: Focus interventions on high-risk states like Kerala
- Use models for annual risk forecasting and resource allocation

8 Limitations and Future Work

8.1 Limitations

- Strong reliance on Accidents_prev — may not capture sudden changes
- No external features (vehicles registered, rainfall, enforcement)
- Limited years (2016–2022) — short time series
- State-level aggregation hides district variation

8.2 Future Work

- Add vehicle count, weather, and policy data
- Use time-series models (ARIMA, LSTM)
- Predict at district level
- Build interactive dashboard for real-time forecasting

9 Conclusion

This project demonstrates that machine learning can effectively predict road accident numbers and classify high-risk state-years using only historical accident data and rates. The strong temporal persistence (previous year accidents dominate) allows high accuracy but highlights the need for proactive interventions to break the cycle in high-risk states like Kerala.

The models provide a foundation for data-driven road safety planning and early warning systems.